

Appendix 3 - Topic modelling

Table 3A - Topic modelling results from BSRLM corpus (N=69)

Topic	Percent	Label	Top_20_Prob	Top_20_FREX
Topic 1	25.39	Attainment, Assessment and Classroom Practices	students, gender, school, gender differences, performance, pupils, national, boys, practices, learning, girls, knowledge, assessment, numeracy, interviews, science, outcomes, undergraduate, views, written	numeracy, gender differences, practices, national, pupils, performance, questions, PGCE, written, problem, knowledge, strategy, location, assessment, curricula, division, students' views, outcomes, calculation, ICT
Topic 2	19.7	Educational pathways, Equity and Policy	attainment, education, gender, participation, England, girls, STEM, attitudes, secondary, social, grouping, factors, gendered, equity, ethnicity, GCSE, policy, post-16, key stage, outcomes	STEM, attainment, participation, grouping, policy, post-16, equity, ethnicity, core, funded, England, gendered, construct, emerging, linguistic, TIMSS, key stage, large-scale, engineering, post-compulsory
Topic 3	19.34	Student Affect, Anxiety and self-efficacy	anxiety, children, girls, secondary, boys, gender, school, rural, students, course, age, self-efficacy, Bangladesh, perceptions, learners, optional, parents, achievement, interactions, support	Bangladesh, rural secondary, anxiety, children, optional course, parents, score, higher-mathematics optional course, self-efficacy, secondary school, feelings, anxiety levels, word problems, adults, learners, perceptions, parental, mixed, studying, mental
Topic 4	17.93	Mathematical Identity and discourses	gender, achievement, girls, discursive, student, variables, age, cultural capital, drawing, geek, identity, mathematics-related, social, participation, teacher, performance, culture, context, Ireland, choosing	choosing, discourses, identity, mathematics-related, maturity, constructions, Northern Ireland, represented, discursive, achievement, variables, drawing, cultural capital, geek, belief, masculinity, gender, interest, culture, multilevel
Topic 5	17.63	Teacher education, professional learning and representations	teachers, women, professional learning, teaching, experiences, England, education, beliefs, mathematicians, images, programme, primary, Chinese, careers, stories, women teachers, schools, initial, confidence, gender	professional learning, women, teachers, careers, stories, teachers' experiences, women teachers, professional, experiences, number sense, films, presentation, mathematicians, programme, images, beliefs, Chinese, teaching experience, life, semi-structured

Table 3B - Topic modelling results from the wider mathematics education corpus (N=568)

Topic	Percent	Label	Top_20_Prob	Top_20_FREX
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Topic 1	15.88	Equity & Mathematical identities	students, gender, education, women, identity, participation, classroom, support, social context, equity, experiences, learning, school, university, inclusion, race, feminist perspectives, diversity, mathematical identity, marginalisation	mathematicians, mathematical identity, discourse, equity, marginalisation, political issues, inclusion, women, textbooks, diversity, race, methodology, feminist perspectives, inequality, queer perspectives, social context, equity work, participation, university, classroom
Topic 2	15.61	Gender Differences in Achievement and assessment	gender differences, girls, boys, achievement, performance, school, reasoning, grade level, ability, tests, assessment, significant effects, boy-girl comparisons, explanatory factors, relationships, cognitive ability, socioeconomic status, TIMSS, international comparisons, risk-taking	TIMSS, boy-girl comparisons, gender differences, boys, statistical reasoning, international science, test items, international trends, competition, country comparisons, girl-boy comparisons, achievement differences, risk-taking, attribution, socioeconomic status, cognitive ability, girls, achievement, performance, assessment
Topic 3	13.9	Attitudes, Achievement & Statistics education	student attitudes, achievement, school, attitudes toward statistics, statistical learning, secondary school, gender differences, student achievement, modelling, relationships, grade level, high school, academic achievement, positive attitudes, achievement tests, coeducation, non-cognitive factors, statistics anxiety, school students, attitude measures	attitudes toward statistics, statistics anxiety, student attitudes, student achievement, junior secondary school, coeducation, non-cognitive factors, academic achievement, positive attitudes, achievement tests, sex differences, school students, attitude measures, statistics education, secondary students, gender differences, modelling, relationship, effect, achievement
Topic 4	13.2	Problem Solving, Reasoning & Performance	problem solving, student performance, statistical reasoning, school level, year group, statistics, courses, skills, gender differences, male students, female students, thinking, understanding, grade level, tasks, visualisation, number sense, metacognition, prior knowledge, problem-solving strategies	problem solving, solving problems, graphs, difficulties, errors, prior knowledge, problem-solving strategies, thinking, student performance, introductory courses, operations, visual reasoning, tasks, frequency, number sense, metacognition, statistical reasoning, mathematical understanding, skills, performance
Topic 5	12.82	Teacher Knowledge, Education & Self-Efficacy	teachers, teaching, teacher education, gender, teacher knowledge, schools, self-efficacy, teaching practices, assessment, technology integration, teacher beliefs, pre-service teachers, teaching experience, primary teachers, science teachers, pedagogical knowledge, elementary teachers,	pre-service teachers, female teachers, teacher self-efficacy, teacher education, school teachers, teacher knowledge, primary teachers, TPACK, science teachers, teaching experience, prospective teachers, teacher practice, elementary teachers, pedagogy, games, technology integration, teacher beliefs,

			prospective teachers, TPACK, professional development	professional development, teaching confidence, assessment practice
Topic 6	10.51	STEM Participation, Interest & Careers	STEM, students, science, school, gender, careers, interest, education, engineering, technology, programmes, female students, participation, science and technology, STEM subjects, girls, stereotypes, STEM education, STEM majors, STEM fields	STEM education, STEM careers, engineering STEM, technology and engineering, STEM majors, science and technology, STEM interest, students in STEM, STEM fields, STEM subjects, biology, occupations, STEM camps, stereotypes, career interest, female participation, engineering, science, technology, school STEM
Topic 7	10.24	Learning Strategies, Confidence & Mathematics Anxiety	learning, mathematics anxiety, learning strategies, gender differences, confidence, achievement, course performance, calculus, scores, grade level, self-regulation, learning styles, emotional factors, blended learning, student learning, preferences, platform use, higher-level courses, final grades, precalculus	learning styles, self-regulation, confidence, learning preferences, emotional factors, learning strategies, blended learning, calculus, mathematics anxiety, student learning, precalculus, higher-level courses, learning platforms, final grades, strategy use, achievement, course performance, gender differences, assessment scores, confidence measures
Topic 8	7.85	Race, Curriculum & Educational Justice	education, girls, Black students, curriculum, perceptions, challenges, concept learning, technology integration, learners, teaching, school level, children, Black women, Black girls, justice, STEAM, African American students, virtual learning, manipulatives, digital learning	Black women, Black girls, STEAM, COVID-19, pupils, curriculum, justice, preschool, dual enrolment, African American students, curriculum challenges, manipulatives, virtual learning, African students, digital learning, concept learning, children, technology integration, perceptions, school curriculum